

Volumes with Cross Sections

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Name: _____ Date: _____ Score: _____

Find the volume of each solid using cross sections. Show all work.

Calculus 1 Worksheet #57 Introduction to Volumes with Cross Sections

Learn: Today you learn how to find volumes using CROSS SECTIONS.

Notes: Volume = $\int$ Area of Cross Sections

1. $\lim_{x \rightarrow \infty} \frac{20x^2 - 13x + 5}{5 - 4x^3} =$

2. $\lim_{h \rightarrow 0} \frac{\ln(2+h) - \ln 2}{h} =$

3. If, $y = e^{-x^2}$, then $y''(0) =$

4. $\int 5^{2x}$

5. $\int_{\pi/4}^{\pi/2} \sin^3 x \cos x \, dx =$

6. $\int_0^1 \frac{e^x}{(3-e^x)^2} \, dx =$ A) $3\ln(e-3)$ B) 1 C) $\frac{1}{3-e}$ D) $\frac{e-2}{3-e}$ E) None of these

7. The average value of $\frac{1}{2}t^2 - \frac{1}{3}t^3$ over the interval $-2 \leq t \leq 1$ is

8. At what value of h is the rate of increase of $\sqrt{h}$ twice the rate of increase of h ?

9. A function $f(x)$ equals $\frac{x^2 - x}{x - 1}$ for all x except $x = 1$. In order that the function be continuous at $x = 1$, the value of $f(1)$ must be?

10. $\int (\sqrt{x} - 2)x^2 \, dx =$

11. If $y = \frac{x-3}{2-5x}$, then $\frac{dy}{dx} =$

12. The hypotenuse AB of a right triangle ABC is 5 ft, and one leg, AC, is decreasing at the rate of 2 ft/sec. The rate, in square feet per second, at which the area is changing when $AC = 3$ is

13*. Find the volume of the solid whose base is the region bounded between the curves $y = x$ and $y = x^2$, and whose cross sections perpendicular to the x -axis are squares.

14*. The base of a certain solid is the region enclosed by $y = \sqrt{x}$, $y = 0$, and $x = 4$. Every cross section perpendicular to the x -axis is a semicircle with its diameter across the base. Find the volume of the solid.

1. Find the volume of the solid generated when the region enclosed between $y = \sqrt{x}$, $x = 1$, $x = 4$, and the x -axis is revolved about the y -axis.

Answers:



Math is hard. So we made it human.
Labanan ang agwat.

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Volume by Cross Sections Worksheet

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Volumes with Cross Sections

Numberbender | ANSWER KEY



Answer key — for instructor use only.

1) 0	2) $\frac{1}{2}$	3) -2	4) $\frac{5^{2x}}{2\ln 5} + C$	5) $\frac{3}{16}$	6) E
7) $\frac{11}{12}$	8) $\frac{1}{16}$	9) 1	10) $\frac{2}{7}x^{\frac{7}{2}} - \frac{2}{3}x^3 + C$	11) $\frac{-13}{(2-5x)^2}$	12) $-\frac{7}{2}$
13) $\frac{1}{30}$	14) π	15) $\frac{124\pi}{5}$			

